

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (1 Question × 100 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	M Rakesh	Reg. No.:	192421006
Section / Batch:	IT	Date:	23/7/26

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1	100
GRAND TOTAL:		100 Marks	

SECTION A – Technical Presentation

Q1. Visualization-Based Technical Presentation on Growth Functions

Analyze growth functions $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, and $O(2^n)$ using visualization tools in a technical presentation format. Clearly define the purpose of comparing these functions for understanding algorithm efficiency. Use graphs, tables, or charts to show how each function behaves as the input size increases. Explain which functions remain practical for large-scale problems and justify your reasoning. Present the conclusions clearly and professionally by linking growth behavior to algorithm selection.

Code of Conduct

I _____ M Rakesh _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

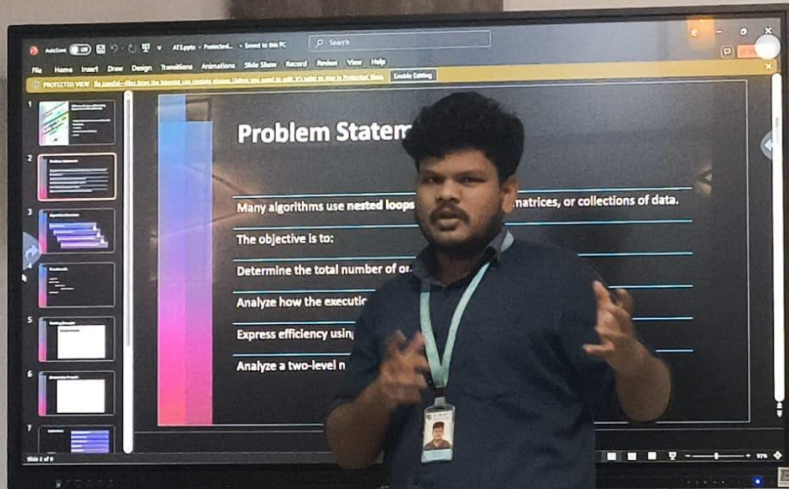
RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1	4	4	4	2	3	90
Overall Total (100)						90

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



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Efficiency Analysis of Counting Operations in Nested Loops

- **Course:** Design and Analysis of Algorithms (DAA)
- **Presented by:**
- M. Rakesh
- B.Tech Information Technology
- SIMATS

Problem Statement

Many algorithms use **nested loops** to process arrays, matrices, or collections of data.

The objective is to:

Determine the total number of operations performed.

Analyze how the execution time grows with input size.

Express efficiency using asymptotic notation.

Analyze a two-level nested loop and derive its time complexity.

Algorithm Overview

Input: Array of size n

Outer loop executes n times.

Inner loop executes n times for every outer iteration.

Each inner iteration performs one counting operation.

Working Example

Example: $n = 3$

Outer (i)	Inner (j)	Count
1	1	1
1	2	2
1	3	3
2	1	4
2	2	5
2	3	6
3	1	7
3	2	8
3	3	9

Final Count = 9

Formula:

$$\text{Total Operations} = n^2$$

For $n = 3$

$$3^2 = 9$$

Interpretation:

- Outer loop runs 3 times ($i = 1$ to 3)
- For each i , inner loop runs 3 times ($j = 1$ to 3)
- Total operations = $3 \times 3 = 9$
- Hence, count = n^2

Complexity Analysis

◆ Number of Operations

• Outer Loop:

Runs from $i = 1$ to $n \rightarrow n$ iterations

• Inner Loop:

For each i , runs from $j = 1$ to $n \rightarrow n$ iterations

Total Number of Operations

For each of the n iterations of the outer loop, the inner loop performs n operations.

$$T(n) = n \times n$$

$$T(n) = n^2$$

Breakdown

If the basic operation inside the inner loop takes c time, then total time:

$$T(n) = c \times n^2$$

As c is a constant, we ignore it in asymptotic notation.

◆ Asymptotic Complexity

Best Case

$$O(n^2)$$

Both loops execute completely.

Average Case

$$O(n^2)$$

Both loops execute completely.

Worst Case

$$O(n^2)$$

Both loops execute completely.

Why $O(n^2)$?

- Outer loop runs n times.
- For each iteration of outer loop, inner loop runs n times.
- Total operations = $n \times n = n^2$
- As n grows larger, time grows proportionally to n^2 .

◆ Space Complexity

$$O(1)$$

Only a few variables (i, j, count) are used regardless of input size n .

★ **Conclusion:** The algorithm performs n^2 basic operations.

Time Complexity = $O(n^2)$, Space Complexity = $O(1)$

Applications

Matrix Multiplication

Image Processing

Pattern Searching

Bubble Sort

Selection Sort

Graph Algorithms (Adjacency Matrix)

Distance Calculations

Results & Discussion

Results

Operation Count

n	Operations (n^2)
5	25
10	100
20	400
50	2500
100	10000

Discussion

- Operations increase quadratically.
- Doubling input size nearly quadruples the work.
- Efficient for small datasets.
- Less suitable for very large inputs.
- Complexity analysis helps choose scalable algorithms.